Figure 3: A fluid is contained between two pistons and partitioned by a transition zone, depicted by the hatched disk in the center. On either side of the transition, the fluid properties are uniform. As the fluid passes through the transition, its properties change smoothly from the initial state to the final state, indicated by subscripts 0 and 1, respectively. The flow is purely one-dimensional and along the line parallel to the unit vector $\hat{\boldsymbol{n}}$.

A Shock Jump Equations

An idealized shock wave is a propagating discontinuity in the variables that describe the fluid. Here we use a simple physical picture to derive the relationships between the fluid properties in front of and behind a shock. We will not be concerned with the development of a shock from an acoustic disturbance or with the microscopic details of the shock transition itself. These topics are addressed in the classic texts by Courant & Friedrichs (1976), Liepmann & Roshko (2001), Landau & Lifshitz (2004), and Zel'dovich & Raizer (2002).

Consider the situation illustrated in Fig. 3. A fluid is enclosed in a cylindrical vessel with unit cross sectional area that is capped on the left and right by moving pistons. Both pistons move to the right—in the direction of $\hat{\boldsymbol{n}}$—but at different speeds. The hatched disk in Fig. 3 is a transition zone, assumed to be at rest and parallel to the pistons. On either side of the transition, the fluid pressure p, specific volume V, internal energy e, and flow speed u are assumed to be spatially uniform and constant in time. Inside the transition zone, the fluid properties change smoothly from state 0 to state 1 according to a set of partial differential equations. However, the details of the transition are not required if our main interest is in the two uniform regions. We need only appeal to the conservation of mass, momentum, and energy to obtain a set of algebraic equations that relate states 0 and 1. These equations are often called the *Rankine-Hugoniot equations* or simply the *jump equations*.

Three conservation laws yield three independent equations for the four final-state variables (V_1, p_1, e_1, u_1), where we assume that the initial conditions (V_0, p_0, e_0, u_0) are known. A fourth independent equation is required to obtain a unique solution for the final state of the fluid. This additional constraint is usually provided as a thermodynamic equation of state of the form $e(p, V)$ or $p(e, V)$, although there are other possibilities for the closure relation, which we discuss elsewhere in this report.

A.1 Mass Conservation

Fluid is pushed by the left piston into the transition zone with speed u_0 and emerges on the other side with speed u_1. In a time interval Δt, the mass[2] of fluid entering the transition is $(u_0/V_0)\Delta t$. Similarly, the mass exiting the transition in time Δt is $(u_1/V_1)\Delta t$. Because the conditions on either side of the transition are assumed to be stationary, the masses entering and exiting the transition in time Δt must be equal, so that

$$\frac{u_0}{V_0} = \frac{u_1}{V_1} \equiv \mu \ , \tag{A1}$$

where we have introduced the symbol μ for the conserved mass flux. Notice that this equation mathematically expresses the intuitive idea that the fluid is compressed $(V_1/V_0 < 1)$ if it is forced to decelerate $(u_1/u_0 < 1)$ across the transition[3].

A.2 Momentum Conservation

Any net change in the momentum of the enclosed fluid must be attributed to unequal forces on the pistons. The left and right pistons exert respective forces of $p_0\hat{\boldsymbol{n}}$ and $-p_1\hat{\boldsymbol{n}}$, so that the net force on the fluid system is $(p_0-p_1)\hat{\boldsymbol{x}}$. In a time interval Δt, the momentum delivered to the fluid by external forces is $(p_0-p_1)\Delta t\hat{\boldsymbol{n}}$. Over the same time span, a mass of $(u_0/V_0)\Delta t$ is pushed into the transition from the left, and the corresponding change in momentum of the left region is $-(u_0^2/V_0)\Delta t\hat{\boldsymbol{n}}$. Similarly, a momentum $(u_1^2/V_1)\Delta t\hat{\boldsymbol{n}}$ is added to

[2]This should say "mass per unit area," but recall that we have set the cross sectional area to 1.

[3]Consider the flow of traffic on a one-lane highway when one vehicle, assumed here to be the lead vehicle, brakes suddenly. There is, in general, some delay in the response of the second vehicle to the braking of the first. Consequently, the distance between the first and second vehicles decreases as the second vehicle attempts to match the speed of the first. The third vehicle responds in turn, and so on down the line. In this fashion, a wave travels backward relative to the flow of traffic, wherein the mean separation between vehicles and the mean vehicle speed decreases as the wave passes. See Lighthill & Whitham (1955) for a mathematical analysis of this problem.

the right of the transition in time Δt. Equating the the net momentum delivered to the fluid by the pistons and the net change in the momentum of the fluid as it crosses the transition, we find

$$p_1 + \frac{u_1^2}{V_1} = p_0 + \frac{u_0^2}{V_0} \ . \tag{A2}$$

A.3 Energy Conservation

Note that eqs. (A1), (A2), (A6), and (A5) result from purely mechanical considerations and make no reference to the microscopic nature or thermodynamic properties of the fluid. It is when we consider the energy of the fluid that we must be aware of the distinct forms the energy can take, such as the translation, rotation, and vibration of individual molecules, the potential energy associated with intermolecular forces, and the energy of the bulk flow. Fortunately, we can overlook these complexities for now and partition the energy into two packages: the internal energy per unit mass e, which encompasses all forms of molecular motions and the intermolecular potential, and the kinetic energy per unit mass $u^2/2$ of the flow. Therefore, the total energy per unit mass of the fluid is $e + u^2/2$.

Here we neglect the addition or subtraction of heat that accompanies chemical reactions or phase changes, and we assume that there is no exchange of heat between the fluid and the external environment[4]. Under these assumptions, any net difference in the energy of the fluid across the transition must be due to work done by the pistons. In general, the rate of work is the dot product of the applied force and the velocity. Therefore, in time Δt, the work done on the fluid by the left an right pistons are, respectively, $p_0 u_0 \Delta t$ and $-p_1 u_1 \Delta t$, and the total work is $(p_0 u_0 - p_1 u_1)\Delta t$. As fluid moves through the transition in time Δt, the changes in the energy on the left and right sides are, respectively, $-(u_0/V_0)(e_0 + u_0^2/2)\Delta t$ and $(u_1/V_1)(e_1 + u_1^2/2)\Delta t$. Equating the work done by the pistons and the total change in fluid energy, we obtain

$$\frac{u_1}{V_1}\left(e_1 + \frac{1}{2}u_1^2\right) + p_1 u_1 = \frac{u_0}{V_0}\left(e_0 + \frac{1}{2}u_0^2\right) + p_0 u_0 \tag{A3}$$

We know from the conservation of mass flux (eq. [A1]) that $u_0/V_0 = u_1/V_1$, which allows us to rewrite the energy conservation condition as

$$e_1 + p_1 V_1 + \frac{1}{2}u_1^2 = e_0 + p_0 V_0 + \frac{1}{2}u_0^2 \ . \tag{A4}$$

A.4 Auxiliary Equations and the Hugoniot Curve

The conservation of mass, momentum, and energy have provided us with three equations (eqs. [A1], [A2], and [A4]) that relate the four state variables (V, p, e, u) on either side of the transition. From these equations, we can derive a number of useful auxiliary relations. For instance, we can combine the mass and momentum jump equations in several different ways:

$$p_1 - p_0 = \mu^2(V_0 - V_1) \ , \tag{A5}$$

$$p_1 - p_0 = \mu(u_0 - u_1) \ , \tag{A6}$$

$$(p_1 - p_0)(V_0 - V_1) = (u_0 - u_1)^2 \ . \tag{A7}$$

The first of these equations, sometimes called the *Rayleigh relation*, is often useful when performing algebraic manipulations of the jump equations. Note that the function $p_1(V_1)$ given by eq. (A5) is a straight line with slope $-\mu^2$. Equations (A6) and (A7) prove useful in various contexts and reduce to very simple forms in a frame of reference where the upstream fluid is at rest.

A fourth auxiliary relation, and perhaps the most important, is obtained as follows. Use eqs. (A1) and (A5) to solve for u_0 and u_1 in terms of the pressure and volume variables, and substitute the results into the expression (eq. [A4]) for energy conservation to obtain

$$e_1 - e_0 = \frac{1}{2}(p_1 + p_0)(V_0 - V_1) \ . \tag{A8}$$

[4] It is straightforward to include additional sources and sinks of heat while the fluid is in the transition zone (e.g., Landau & Lifshitz 2004, pg. 489–494).

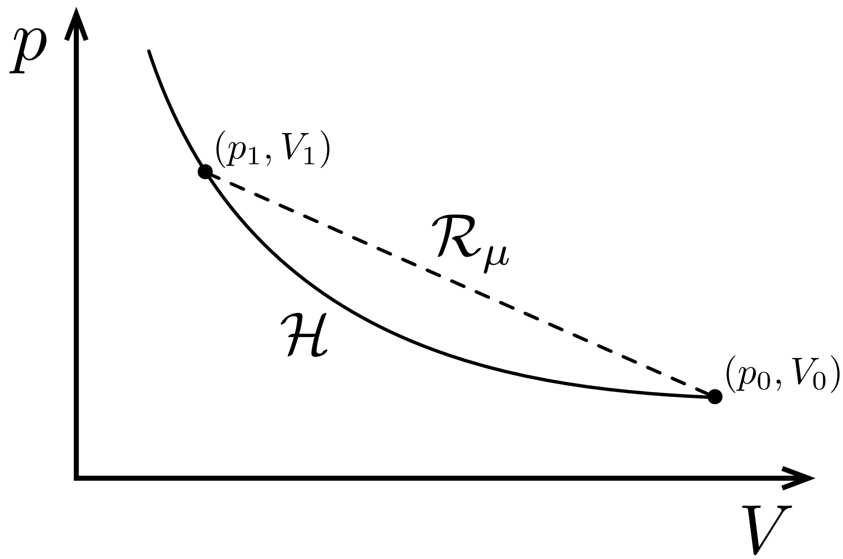

Figure 4: The Hugoniot curve, labeled by $\mathcal{H}$, is the locus of possible fluid states following the passage of a shock through a fluid with initial conditions (p_0, V_0). For a particular final state (p_1, V_1), the Rayleigh line, labeled by $\mathcal{R}_\mu$, results from the combination of the mathematical expressions for mass and momentum conservation (see eqs. [A1], [A2], and [A5]).

If we have an equation of state in the form $e(p, V)$, eq. (A4) yields a function $p_1(V_1; p_0, V_0)$ called the *Hugoniot curve*, where V_1 is the independent variable, and p_0 and V_0 specify the initial conditions. Figure 4 shows an illustrative depiction[5] of the Hugoniot curve, labeled $\mathcal{H}$, in addition to the Rayleigh line, labeled $\mathcal{R}_\mu$ (see eq. [A5]), that connects the initial and final states. After specifying u_0, and thus μ, for a given (p_0, V_0), the final state (p_1, V_1) is the intersection of $\mathcal{H}$ and $\mathcal{R}_\mu$. Equation (A8) is often used in place of eq. (A4) for the energy jump equation.

Of course, the equation of state need not be in the form $e(p, V)$; it can relate any set of the state variables. A different choice for the form of the equation of state results in a different functional relationship that represents the locus of possible post-shock states. When we use the phrase "Hugoniot curve," or some times just "Hugoniot," in this report, we do not necessarily refer to the specific form $p_1(V_1; p_0, V_0)$, but to any representation of the solution space for the post-shock conditions.

A.5 The Lab Frame

It is often the case in laboratory experiments and in other practical situations that the shocked fluid is initially at rest and the shock front is moving. When this is the case, we say that we are working in the *lab frame*. Figure 5 shows a shock wave making an angle α with the horizontal and progressing from right to left with velocity $\boldsymbol{u}_s$. When $\alpha = \pi/2$ the shock is *normal*; otherwise, the shock is *oblique*. As a fluid element passes through the shock, the thermodynamic state changes from (V_0, p_0, e_0) to (V_1, p_1, e_1) and the velocity changes from 0 to $\boldsymbol{u}_f$.

The jump equations for oblique shocks in the lab frame are easily derived by first considering the conditions in the shock rest frame and then carrying out the appropriate Galilean transformations to the frame in which the upstream fluid is at rest. Panel (a) of Fig. 6 shows the most general situation in the rest frame of the shock. Both the upstream and downstream flow velocities have nonzero components normal and tangent to the shock, as denoted by subscripts n and t. A fluid element passing through the shock feels a force in the direction of decreasing pressure, which in this case is the upstream direction. It follows that only the normal component of the flow velocity is affected by the shock, and the tangential component is preserved. In the rest frame of the shock, the jump equations are obtained by replacing u_0 and u_1 in eqs. (A1), (A2),

[5] The exact mathematical representation of the Hugoniot curve obviously depends on the chosen equation of state. Nonetheless, many shocked materials exhibit a curve similar to the one shown in Fig. 4. In particular, for an ideal gas, with $e = (\gamma - 1)^{-1} pV$, the Hugoniot curve is a hyperbola with a vertical asymptote at a specific volume of $(\gamma - 1)/(\gamma + 1)$.

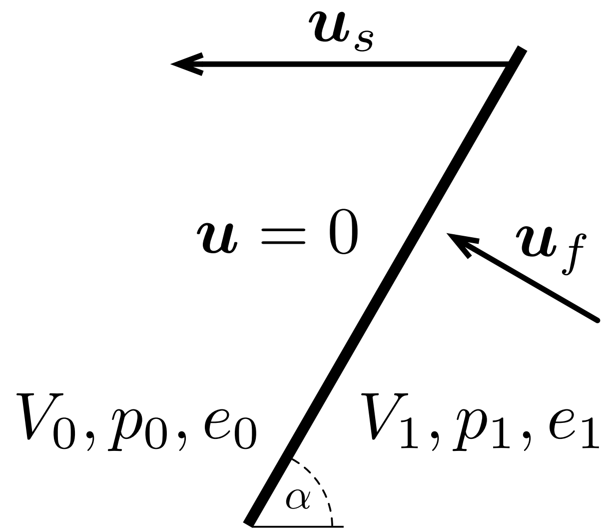

Figure 5: An oblique shock in the lab frame. The shock wave (thick solid line) moves to the left with velocity $\boldsymbol{u}_s$ and makes an angle α with the horizontal. Fluid is brought from rest to a velocity $\boldsymbol{u}_f$ during the passage of the shock.

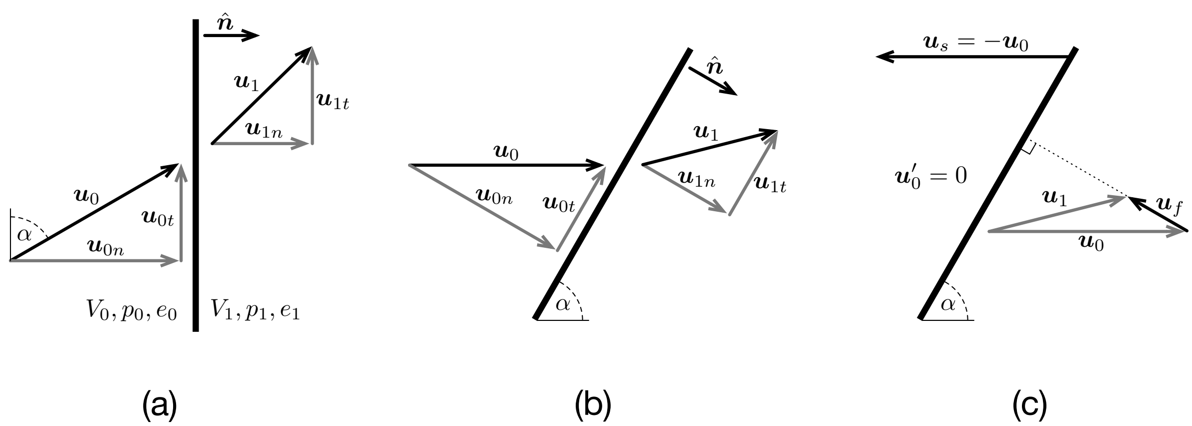

Figure 6: Panel (a) of the above figure depicts the conditions in the rest frame of the shock, where $\hat{\boldsymbol{n}}$ is the shock normal (pointing downstream), and α is the angle between $\boldsymbol{u}_0$ and the front. The upstream and downstream flow velocities are decomposed into components normal and tangent to the front, as denoted by subscripts n and t, respectively. In panel (b), the entire system has been rotated such that $\boldsymbol{u}_0$ points horizontally from left to right and the shock is at an angle α from the horizontal. Panel (c) shows the picture seen by an observer when the oblique shock moves horizontally to the left with velocity $\boldsymbol{u}_s = -\boldsymbol{u}_0$. In this frame, the upstream fluid is at rest ($\boldsymbol{u}_0' = 0$), and the downstream flow velocity is $\boldsymbol{u}_f = \boldsymbol{u}_1 - \boldsymbol{u}_0$.

and (A4) with $\boldsymbol{u}_0 \cdot \hat{\boldsymbol{n}}$ and $\boldsymbol{u}_1 \cdot \hat{\boldsymbol{n}}$, respectively, to yield

$$\begin{aligned}
\boldsymbol{u}_1 \cdot \hat{\boldsymbol{n}}/V_1 &= \boldsymbol{u}_0 \cdot \hat{\boldsymbol{n}}/V_0 \ , \\
p_1 + (\boldsymbol{u}_1 \cdot \hat{\boldsymbol{n}})^2/V_1 &= p_0 + (\boldsymbol{u}_0 \cdot \hat{\boldsymbol{n}})^2/V_0 \ , \\
e_1 + p_1 V_1 + \frac{1}{2}(\boldsymbol{u}_1 \cdot \hat{\boldsymbol{n}})^2 &= e_0 + p_0 V_0 + \frac{1}{2}(\boldsymbol{u}_0 \cdot \hat{\boldsymbol{n}})^2 \ . \qquad \text{(A9)}
\end{aligned}$$

Now that we have the jump equations in a general vector-based form, the transformation to the lab frame is straightforward.

In the shock rest frame, the velocity $\boldsymbol{u}_0$ makes an angle α with the front. In panel (b) of Fig. 6, the shock rest frame has been rotated so that the upstream flow is horizontal and the shock is inclined at an angle $\alpha \in [0, \pi/2]$ to the horizontal. To complete the transformation to the lab frame, add $-\boldsymbol{u}_0$ to all the velocity vectors (Fig. 6, panel c). The velocity of the oblique shock in the lab frame is $\boldsymbol{u}_s = -\boldsymbol{u}_0$, the upstream velocity is $\boldsymbol{u}_0' = \boldsymbol{u}_0 - \boldsymbol{u}_0 = 0$, and the downstream velocity is $\boldsymbol{u}_f = \boldsymbol{u}_1 - \boldsymbol{u}_0$. Since $u_{1t} = u_{0t}$, we see that $\boldsymbol{u}_f$ points normal to the shock. To obtain the transformed jump equations, make the replacements $\boldsymbol{u}_0 = -\boldsymbol{u}_s$ and $\boldsymbol{u}_1 = \boldsymbol{u}_f - \boldsymbol{u}_s$ in eq. (A9):

$$\begin{aligned}
(\boldsymbol{u}_f - \boldsymbol{u}_s) \cdot \hat{\boldsymbol{n}}/V_1 &= -\boldsymbol{u}_s \cdot \hat{\boldsymbol{n}}/V_0 \ , \\
p_1 + [(\boldsymbol{u}_f - \boldsymbol{u}_s) \cdot \hat{\boldsymbol{n}}]^2/V_1 &= p_0 + (\boldsymbol{u}_s \cdot \hat{\boldsymbol{n}})^2/V_0 \ , \\
e_1 + p_1 V_1 + \frac{1}{2}[(\boldsymbol{u}_f - \boldsymbol{u}_s) \cdot \hat{\boldsymbol{n}}]^2 &= e_0 + p_0 V_0 + \frac{1}{2}(\boldsymbol{u}_s \cdot \hat{\boldsymbol{n}})^2 \ . \qquad \text{(A10)}
\end{aligned}$$

The first line above defines the conserved mass flux $\mu = -\boldsymbol{u}_s \cdot \hat{\boldsymbol{n}}/V_0$, which is positive due to our choice of the direction of $\hat{\boldsymbol{n}}$.

With the substitutions $u_{sn} = |\boldsymbol{u}_s \cdot \hat{\boldsymbol{n}}| = u_s \sin\alpha$ and $u_f = |\boldsymbol{u}_f \cdot \hat{\boldsymbol{n}}|$, the mass jump equation can be rewritten as

$$\frac{u_f}{u_{sn}} = 1 - \frac{V_1}{V_0} \equiv \beta \ , \qquad \text{(A11)}$$

where we have introduced the compression parameter $\beta \in (0, 1)$. Auxiliary equations (A6) and (A7) transform to

$$p_1 - p_0 = -\mu u_f = \frac{u_{sn}\, u_f}{V_0} \ , \qquad \text{(A12)}$$

and

$$V_0(p_1 - p_0)\beta = u_f^2 \ . \qquad \text{(A13)}$$

When dealing with shocks of known velocity $\boldsymbol{u}_s$ in the lab frame, eqs. (A11) and (A12) are especially convenient forms of the mass and momentum jump equations. Equation (A13) allows us to write the Hugoniot relation (eq. [A8]) as

$$e_1 - e_0 = \frac{1}{2}u_f^2 + p_0 V_0 \beta \ . \qquad \text{(A14)}$$

A.6 Summary and Approximations

As a summary of the main results of this appendix, we now write the three jump equations in the forms that are most useful in this report. In the lab frame, the mass, momentum, and energy jump equations are

$$\frac{u_f}{u_{sn}} = \beta \ , \qquad \text{(A15)}$$

$$p_1 - p_0 = \frac{u_f\, u_{sn}}{V_0} \ , \qquad \text{(A16)}$$

and

$$
\begin{aligned}
e_1 - e_0 &= \frac{1}{2} V_0 (p_1 + p_0) \beta \\
&= \frac{1}{2} u_f^2 + p_0 V_0 \beta \ .
\end{aligned}
\tag{A17}
$$

where we have used the Hugoniot relation (eq. [A8]) for the energy condition. These three equation for the four unknown final-state parameters are completely general and make no reference to the microphysical properties of the fluid. In order to close the system of equations, an additional constraint must be provided, such as a thermodynamics equation of state (e.g., $e(p, V)$ or $p(e, V)$).

Certain simplifications arise when $p_1 \gg p_0$ and $e_1 \gg e_0$ (the *strong shock* conditions). In particular, the momentum and energy jump equation become

$$
p_1 \approx \frac{u_{sn} u_f}{V_0} \tag{A18}
$$

and

$$
\begin{aligned}
e_1 &\approx \frac{1}{2} V_0 p_1 \beta \\
&\approx \frac{1}{2} u_f^2 \ .
\end{aligned}
\tag{A19}
$$

In this limit, the total specific energy added to the fluid as it crosses the transition, $e_1 - e_0 + u_f^2/2$, is roughly equipartitioned between the kinetic energy of the bulk flow and the internal energy.

B Equations of State for Shocked Condensed Matter

There is a great deal of literature on liquid and solid equations of state. Rather than provide a comprehensive review, we will highlight only a few useful results that are applicable to shocked substances at pressures below ∼50 GPa. For more thorough discussions of the equations of state of shock-loaded condensed materials, see Harvey (1986), Segletes (1991), Kerrisk & Harvey (1997), Zel'dovich & Raizer (2002, Part XI), and additional articles cited below.

Near the initial state, the change in specific entropy across a shock is, to lowest order, $s - s_0 \propto \beta^3$ (e.g., Zel'dovich & Raizer 2002, Part I, § 18). The proportionality coefficient is positive for the vast majority of substances, so that entropy increases within the shock transition. A sufficiently weak shock ($\beta \ll 1$) leads to a nearly adiabatic compression of the material. When the pressure is written as a function of V and s, the change in pressure under a general thermodynamic transformation is

$$p - p_0 = (V - V_0)\left(\frac{\partial p}{\partial V}\right)_{s_0} + (s - s_0)\left(\frac{\partial p}{\partial s}\right)_{V_0} + \dots \ . \tag{B1}$$

Since the entropy change is third order along the Hugoniot, the first-order approximation for the change in pressure across the shock is given by the first term in the above expansion. Using the definitions of the bulk modulus K and adiabatic sound speed[6] c,

$$K = -V\left(\frac{\partial p}{\partial V}\right)_s = \frac{c^2}{V} \ , \tag{B2}$$

we see that for small compressions ($\beta = 1 - V/V_0 \ll 1$) the pressure along the Hugoniot is

$$p - p_0 \approx \beta K_0 = \beta \frac{c_0^2}{V_0} \ . \tag{B3}$$

This is essentially just a statement of Hooke's law for small deformations, where the stress $p - p_0$ is proportional to the strain β, and K_0 characterizes the stiffness of the substance. Compare eq. (B3) to the general expression (eq. [A16]) for the pressure jump across a shock, which can be written as

$$p - p_0 = \beta \frac{u_{sn}^2}{V_0} \ . \tag{B4}$$

We see immediately that the shock speed approaches the sound speed as the shock strength is reduced, which must be the case if the Hugoniot and isentrope are tangent at the initial state. Any EoS of the form $p(s, V)$ that we adopt for a shocked liquid must reduce to eq. (B3) as $p - p_0$ approaches 0.

Murnaghan (1944) generalized the linear adiabatic response in eq. (B3) by supposing that K is a linear function of the pressure[7]:

$$K = K_0 + m(p - p_0) \ . \tag{B5}$$

When the dimensionless slope m is positive, this relation has the desired property that the substance stiffens as it is compressed. Murnaghan then solved the differential equation

$$-V\left(\frac{\partial p}{\partial V}\right)_s = K_0 + m(p - p_0) \tag{B6}$$

along the isentrope with entropy s_0 to obtain a nonlinear adiabatic EoS[8],

$$p - p_0 = \frac{K_0}{m}\left[\left(\frac{V_0}{V}\right)^m - 1\right] = \frac{K_0}{m}\left[(1-\beta)^{-m} - 1\right] \ , \tag{B7}$$

[6] The term 'sound speed' is ambiguous for materials with significant rigidity under shear strains. Here we are assuming that either the material has a negligible shear modulus in its initial state, as is true for gases and many liquids, or that the shock is sufficiently strong—much larger than the yield strength—that the substance behaves like a liquid behind the shock. In either case, the propagation speed of an acoustic wave is $c = (KV)^{1/2}$ (see Zel'dovich & Raizer 2002, Part XI, § 16).

[7] The presence of p_0 is our addition; Murnaghan neglected this offset.

[8] This equation is sometimes erroneously referred to as the Tait equation of state, as are a number of other equations derived using similar principles. See Hayward (1967) for a thorough discussion of the Tait equation, including its history, domain of validity, extensions, and incorrect citations in the literature.

which clearly reduces to eq. (B3) in the limit of small β. Although this equation is derived in a somewhat ad hoc fashion and assumes an isentropic transformation, it is convenient mathematically and does provide a good fit (often to within 5–10%) to measured Hugoniot curves up to pressures of $\approx$50 GPa for appropriately chosen values of m. For shocked metals, $m = 4$ is often used, where $K_0 \approx$ 20–50 GPa, and for water and similar liquids, $m \approx$ 7–8 and $K_0 \approx 2$ GPa (e.g., Zel'dovich & Raizer 2002, Part XI, § 9).

Most liquids and solids exhibit a Hugoniot curve that has a simple mathematical representation in terms of the normal shock speed u_{sn} and the post-shock flow speed u_f, both measured in the frame where the fluid is initially at rest. At shock pressures above 2–3 GPa, but not near phase transitions, the data is often well fit by the linear function

$$\frac{u_{sn}}{c_0} = a_1 + a_2 \frac{u_f}{c_0} \ , \tag{B8}$$

where $a_0 \gtrsim 1$ and $a_1 \approx$ 1.5–2 for many liquids (e.g., Duvall & Fowles 1963). Swan (1971) uses the Debye model to provide a theoretical basis for the wide validity of this linear relation in the case of solids. As the shock pressure is decreased below 2–3 GPa, the Hugoniot deviates from the linear fit in order to satisfy the boundary condition that $u_{sn}/c_0 = 1$ when $u_f = 0$. To capture the acoustic limit at low pressures and the asymptotic linear trend for strong shocks, Woolfolk et al. (1973) added to eq. (B8) a function that interpolates between the two regimes:

$$\frac{u_{sn}}{c_0} = a_1 + a_2 \frac{u_f}{c_0} - (a_0 - 1)e^{-a_3 u_f/c_0} \ . \tag{B9}$$

Woolfolk et al. (1973) found that values of $a_1 = 1.37$, $a_2 = 1.62$, and $a_3 = 2$ are consistent with experimental data on water, glycerin, and carbon tetrachloride. Shock data obtained by Mitchell & Nellis (1982) for water and ammonia up to pressures $>$100 GPa indicate $a_1 \approx 1.6$ and $a_2 \approx$ 1.33–1.4 for these two liquids. When reference values are required for the a_i, we will use rational numbers appropriate for water: $a_1 = 5/3$, $a_2 = 4/3$, and $a_3 = 2$.

Equations (B7), (B8), and (B9) are designed to fit the Hugoniot curves of shocked materials; they are not general equations of state. Since the present study is focused on the state of the material immediately behind the shock, the lack of generality of these relations is of little concern. Of greater relevance is the fact that these equations fail to provide a good fit when the substance undergoes a phase transition. A phase transition is signaled by a discontinuity in the slope of the Hugoniot curve. However, the analyses here are intended only to guide our understanding, rather than provide a completely general, rigorous framework. Moreover, it is not feasible to consider a wide variety of specific materials, each with its own detailed response to strong shock-loading. Our hope is that the above equations of state are a sufficient starting point to provide reasonable quantitative estimates of p, V, and e following the passage of a shock.

C Shock Temperature

Thermal energy is composed of the translational and internal motions of the molecules that make up the bulk mass of a substance, as well as free electrons if the temperature is sufficiently high. In this Appendix, we derive an expression that relates the temperature along the Hugoniot curve to the shock speed. We tacitly assume that the substance remains pure as it passes through the shock wave; that is, we neglect phase changes and chemical reactions that release or absorb heat or change the molecular properties of the fluid.

After presenting some general results from basic thermodynamics, we derive a differential equation for the temperature of the shocked fluid. An approximate solution of this differential equation is obtained by making certain simplifying assumptions and using the linear $u_{sn}(u_f)$ shock EoS discussed in Appendix B.

C.1 General Thermodynamics

Begin by recalling the First Law of thermodynamics:

$$dq = de + pdV \ , \tag{C1}$$

where dq is the infinitesimal change in heat per unit mass. Now suppose that the heat is a function of the temperature T and specific volume V, so that

$$dq(T, V) = \left(\frac{\partial q}{\partial V}\right)_T dV + c_V dT \ , \tag{C2}$$

where we have used the definition of the specific heat at constant volume, $c_V = (\partial q/\partial T)_V$. The temperature change is obtained by integrating the differential relation

$$\begin{aligned} dT &= \frac{dq}{c_V} - \frac{V}{c_V}\left(\frac{\partial q}{\partial V}\right)_T \frac{dV}{V} \\ &= \frac{dq}{c_V} - T\,\Gamma\frac{dV}{V} \ , \end{aligned} \tag{C3}$$

where we have introduced the *Grüneisen parameter* $\Gamma = (V/c_V T)(\partial q/\partial V)_T$. While we could have written the differential equation for the temperature in a number of different ways, eq. (C3) is the most convenient form when the integration is along the Hugoniot curve, where changes in V, p, and e, and thus q, are obtained from the shock jump equations and the equation of state.

It is evident from eq. (C3) that the Grüneisen parameter can also be defined as

$$\Gamma = -\left(\frac{\partial \ln T}{\partial \ln V}\right)_q \ , \tag{C4}$$

which is generally expected to be positive. Under an adiabatic transformation ($dq = 0$), we have simply $dT/T = -\Gamma\, dV/V$, so that the temperature increases under a compressive load when $\Gamma > 0$. See Slater (1939) and Segletes (1991) for discussions and interpretations of Γ, including alternative thermodynamic representations and theoretical evaluations. Here we treat Γ as a purely empirical quantity. For many liquids and solids $\Gamma \approx 0.5$–2 over a wide range of volumes, pressures, and temperatures (e.g., Slater 1939; Gurtman et al. 1971; Boehler & Kennedy 1977; Winey et al. 2000).

C.2 Temperature Along the Hugoniot

Equation (C3) is a general relation between thermodynamic derivatives and makes no reference to shocks or a particular EoS. In order to evaluate the post-shock temperature, we need to provide an EoS and integrate eq. (C3) along the shock Hugoniot from the initial thermodynamic state (V_0, p_0, e_0).

When the shock velocity is known and the jump equations (Appendix A) are supplemented by a closure relation of the form $u_{sn}(u_f)$, it is sensible to cast all quantities in terms of the independent variable u_{sn}. To this end, we first note that the change in heat along the Hugoniot is governed by the First Law (eq. [C1]), so that

$$dq_{\mathcal{H}} = de_{\mathcal{H}} - V_0 p\, d\beta_{\mathcal{H}} \ , \tag{C5}$$

where the subscript $\mathcal{H}$ indicates a directional derivative, and the compression parameter β (eq. [A11]) is used in place of V. From the mass jump equation (eq. [A15]), the change in β along the Hugoniot is

$$d\beta_{\mathcal{H}} = -V_0^{-1} dV_{\mathcal{H}} = \left(\frac{du_f}{du_{sn}} - \frac{u_f}{u_{sn}} \right) \frac{du_{sn}}{u_{sn}} \ . \tag{C6}$$

Using the energy jump equation (eq. [A17]), we find

$$de_{\mathcal{H}} = u_f du_f + p_0 V_0 d\beta_{\mathcal{H}} \ . \tag{C7}$$

Equations (C7), (C6), and the momentum jump equation (eq. [A16]) combine to yield the simple expression

$$dq_{\mathcal{H}} = u_f^2 \frac{du_{sn}}{u_{sn}} \ . \tag{C8}$$

Now, the post-shock temperature is obtained by integrating the first-order differential equation

$$dT_{\mathcal{H}} = \left[\frac{u_f^2}{c_V} + T\,\Gamma \left(1 - \frac{u_f}{u_{sn}} \right)^{-1} \left(\frac{du_f}{du_{sn}} - \frac{u_f}{u_{sn}} \right) \right] \frac{du_{sn}}{u_{sn}} \ . \tag{C9}$$

In order to solve this equation, we require functional forms for the specific heat $c_V(u_{sn}, T)$, Grüneisen parameter $\Gamma(u_{sn}, T)$, and the shock EoS $u_{sn}(u_f)$.

C.3 Approximate Results

As a first approximation that hides the material-dependent and theoretical aspects of c_V and Γ, we simply assume that c_V and Γ are constant in eq. (C9). In the specific case of shocked liquid water initially at standard temperature and pressure, this seems quantitatively reasonable, since neither c_V nor Γ vary by more than ≈50% up to shock pressures of 50 GPa (Gurtman et al. 1971; Mitchell & Nellis 1982; Ree 1982). It seems that this approximation is also suitable for other liquids (e.g., Boehler & Kennedy 1977; Winey et al. 2000; Root & Gupta 2009) as long as appropriate mean values of c_V and Γ are chosen such that the calculated temperatures are in rough agreement with the measured values.

When c_V and Γ are fixed, the differential equation for the temperature is *linear*; i.e., eq. (C9) can be written as

$$\frac{dT}{du_{sn}} + F(u_{sn})\,T = G(u_{sn}) \ , \tag{C10}$$

where

$$F(u_{sn}) = -\Gamma \frac{1}{u_{sn}} \left(1 - \frac{u_f}{u_{sn}} \right)^{-1} \left(\frac{du_f}{du_{sn}} - \frac{u_f}{u_{sn}} \right) \ , \tag{C11}$$

and

$$G(u_{sn}) = \frac{1}{c_V} \frac{u_f^2}{u_{sn}} \ . \tag{C12}$$

Equation (C10) has the formal solution

$$T(u_{sn}) = e^{-A(u_{sn})} \left[T_0 + \int_b^{u_{sn}} dx\, G(x)\, e^{A(x)} \right] \ , \tag{C13}$$

where $A(u) = \int_b^u dx F(x)$, x and x' are dummy integration variables, and T_0 is the pre-shock temperature.

Suppose that the shock equation of state $u_{sn}(u_f)$ has the linear form discussed in Appendix B:

$$u_{sn} = b + a u_f \ . \tag{C14}$$

This is also an approximation when a and b are constant (see the discussion in Appendix B). Upon changing the independent variable from u_{sn} to $\tilde{M} = u_{sn}/b$ in eq. (C10), the functions F and G in eqs. (C11) and (C12) are replaced by

$$bF(\tilde{M}) = -\Gamma \frac{1}{\tilde{M}[1 + (a-1)\tilde{M}]} \tag{C15}$$

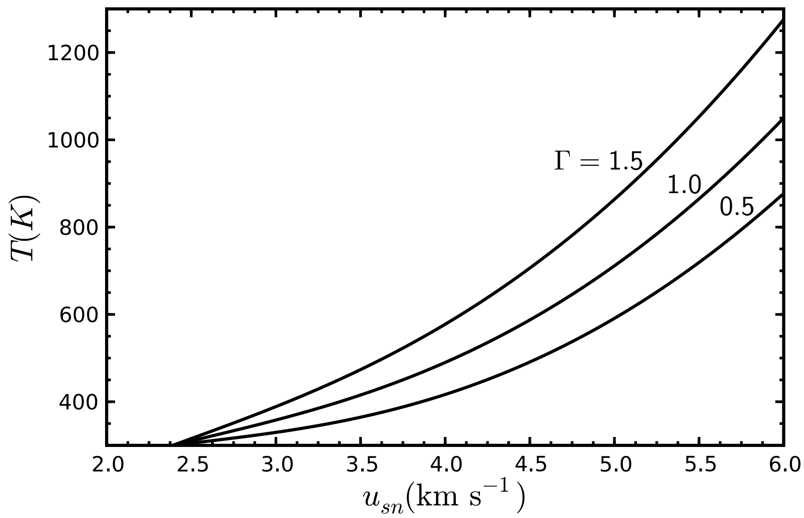

Figure 7: Post-shock temperature calculated from eq. (C18) with $T_0 = 300\,\mathrm{K}$, $a = 1.33$, $b = 2.39\,\mathrm{km\,s^{-1}}$, and $c_V = 4000\,\mathrm{J\,kg^{-1}\,K^{-1}}$. From bottom to top, the three curves correspond to $\Gamma = 0.5$, 1.0, and 1.5.

and

$$bG(\tilde{M}) = \frac{b^2}{a^2 c_V} \frac{(\tilde{M}-1)^2}{\tilde{M}} \ . \tag{C16}$$

When Γ is constant, bF is easily integrated up to some $\tilde{M}$:

$$\exp\left\{-\int_1^{\tilde{M}} dx\, bF(x)\right\} = \left[\frac{a\tilde{M}}{(a-1)\tilde{M}+1}\right]^{\Gamma} \ . \tag{C17}$$

As $\tilde{M}$ varies from 1 to ∞, this function increases monotonically from 1 to $(1-a^{-1})^{-\Gamma}$ for $a > 1$ (see Appendix B). With c_V fixed, the post-shock temperature is, from eq. (C13),

$$T(\tilde{M}) = \left[\frac{a\tilde{M}}{(a-1)\tilde{M}+1}\right]^{\Gamma} \left\{T_0 + \frac{b^2}{a^2 c_V}\int_1^{\tilde{M}} dx\, \left[\frac{(x-1)^2}{x}\right]\left[\frac{(a-1)x+1}{ax}\right]^{\Gamma}\right\} \ . \tag{C18}$$

Notice that b^2/c_V has units of temperature and has a value of order $10^3\,\mathrm{K}$ for most liquids.

There are number of ways to place analytic bounds on the integral in eq. (C18) and approximate the temperature. However, this exercise has limited value, and so we simply integrate numerically and show the results graphically. As an illustrative example, we choose parameters appropriate for water at high shock pressures (e.g., Mitchell & Nellis 1982): $a = 1.33$, $b = 2.39\,\mathrm{km\,s^{-1}}$, $c_V = 4000\,\mathrm{J\,kg^{-1}\,K^{-1}}$, and $\Gamma = 0.5$–1.5. The resulting post-shock temperature is shown in Fig. 7 for $T_0 = 300\,\mathrm{K}$ and Γ drawn from the set $\{0.5, 1.0, 1.5\}$. These curves are in rough agreement with the data in Lyzenga et al. (1982) on shocked water. Up to shock speeds of $5\,\mathrm{km\,s^{-1}}$, the temperature is less than $\approx$1500 K for the range of Γ shown in Fig. 7, and thus the characteristic thermal energy is $kT \lesssim 0.1\,\mathrm{eV}$.

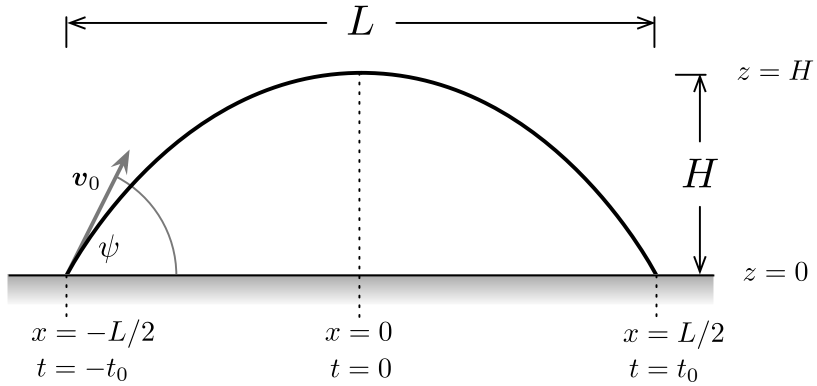

Figure 8: A projectile launched with velocity $\boldsymbol{v_0}$ at an angle ψ to the horizontal moves along a parabolic trajectory over flat ground under constant vertical gravitational acceleration. The trajectory has range L, maximum altitude H, and flight duration $2t_0$.

D Ballistic Flight

In this Appendix, we derive useful formulae relevant to the trajectories of short- and medium-range ballistic missiles. A ballistic trajectory is uniquely specified by two parameters, which could be the initial vertical and horizontal components of the velocity, the initial speed and launch angle, the range and maximum altitude, or any other pairing of these variables. Here we choose the range L and maximum altitude H as our independent parameters, since one or both of these is often quoted for a given missile. Our analysis makes a number of simplifying assumptions: (1) we restrict the range and altitude to relatively small values, (2) air drag is neglected, (3) the Earth's rotation are neglected, and (4) we do not consider thrust or maneuvering after the launch. For a more comprehensive discussion of the dynamics of ballistic missiles, see, e.g., Bate et al. (1971).

For ballistic missiles with ranges much less than the circumference of the Earth and maximum altitudes much less than the Earth's radius, we may approximate the dynamics by assuming flat ground and using a constant gravitational acceleration $g \approx 9.8\,\mathrm{m\,s^{-2}}$. Under these conditions, the trajectory is a parabola determined by the solution of

$$\frac{d\boldsymbol{v}}{dt} = -g\hat{\boldsymbol{z}} \ . \tag{D1}$$

We see immediately that $dv_x/dt = 0$ and that v_x is a constant of the motion. For the velocity in the z direction, we have

$$v_z(t) = -gt \ , \tag{D2}$$

where time $t = 0$ occurs at the top of the parabola. Furthermore, let $-t_0$ and $+t_0$ be the times at launch and impact, respectively. The solution for $[x(t),\ z(t)]$ is

$$[x(t),\ z(t)] = [v_x t,\ H - \frac{1}{2}gt^2] \ , \tag{D3}$$

where $x = 0$ is at the midpoint of the trajectory. The range is given by

$$L = 2v_x t_0 \ , \tag{D4}$$

and $z(t)$ yields

$$t_0^2 = 2H/g \ . \tag{D5}$$

The components of the launch velocity are given by

$$v_x^2 = \frac{gL^2}{8H} \ , \tag{D6}$$

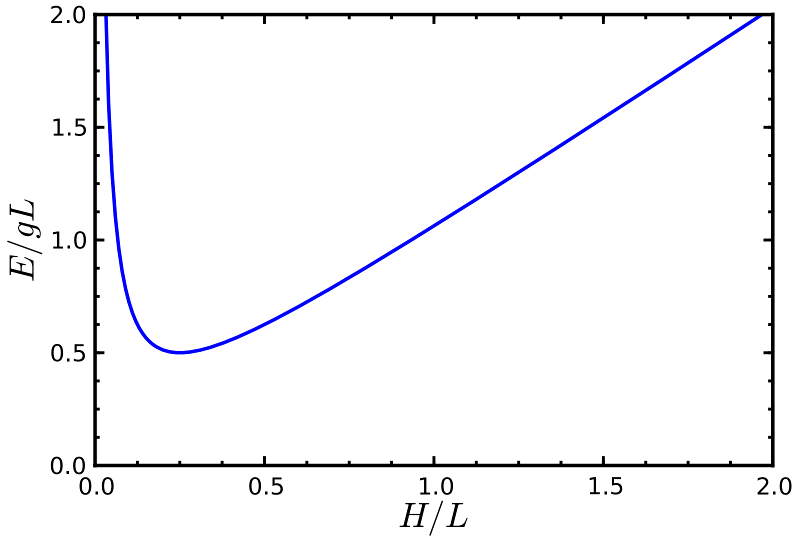

Figure 9: Scaled trajectory energy E/gL as a function of H/L for fixed range L. There are two values of the maximum altitude H with the same energy and range, and that there is a minimum energy at fixed range.

and

$$v_{z0}^2 = 2gH \ . \tag{D7}$$

The launch angle is computed using

$$\tan\psi = \frac{v_{z0}}{v_x} = 4\frac{H}{L} \ . \tag{D8}$$

The conserved energy per unit mass along the parabola is

$$E = \frac{1}{2}v^2 + gz = \frac{1}{2}v_x^2 + \frac{1}{2}v_z^2 + gz \ . \tag{D9}$$

Evaluating the energy at $z = 0$ and using eqs. (D6) and (D7), we obtain

$$E = gH\left[1 + \left(\frac{L}{4H}\right)^2\right] \ . \tag{D10}$$

If L is fixed and H is allowed to vary, the energy initially decreases as H^{-1} for small H, reaches a minimum, and then asymptotically approaches the line $E = gH$ (see Fig. 9). At the minimum energy, where $\partial E/\partial H = 0$, the range and maximum altitude satisfy $H = L/4$, and the value of the minimum energy is given by

$$E_{\min} = 2gH = \frac{1}{2}gL \ . \tag{D11}$$

This is a simple, useful result if it is known that the missile follows a minimum-energy trajectory and either L or H is provided. Since $H = L/4$ for the minimum-energy trajectory, we see that the launch angle, given by eq. (D8), is 45°, a well known result.

When $E > E_{\min}$ for a given range, there are, in general, two permissible trajectories with different maximum altitudes that bracket the minimum-energy result $L/4$:

$$\frac{H}{L} = \frac{E}{2gL}\left\{1 \pm \left[1 - \left(\frac{gL}{2E}\right)^2\right]^{1/2}\right\} \ . \tag{D12}$$

The higher trajectory is referred to as the "lofted" solution, and the shallow path is said to be "depressed" **(ref)**. A lofted trajectory is used when a steep angle of impact is desired or if obstacles along the line of sight to the target prevent one from using the minimum-energy solution. An advantage of the depressed path is its short travel time (see eq. [D5]).

Along a parabolic flight, the speed of the projectile is given by the general relation $v^2(z) = 2(E - gz)$. If L and H are specified, then E is computed from eq. (D10) and $v(z)$ is completely determined for $0 < z < H$. On the other hand, if L and E are fixed, such as when the target location and the launch speed are known, then the launch angle must be one of two choices from the combination of eqs. (D8) and (D12). A relevant special case is when the projectile follows a minimum-energy trajectory for a given range, where the speed is

$$v(z) = (gL)^{1/2} \left(1 - \frac{z}{2H}\right)^{1/2} . \tag{D13}$$

Here the launch speed is $v_0 = (gL)^{1/2}$ and the minimum speed, found at the apex of the flight, is $(gL/2)^{1/2}$. For a range of $L = 1000\,\mathrm{km}$ and a minimum-energy path, the maximum altitude is $H = 250\,\mathrm{km}$, and the launch and apex speeds are $\approx 3.1\,\mathrm{km\,s^{-1}}$ and $\approx 2.2\,\mathrm{km\,s^{-1}}$, respectively.

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